

Topic 8 of Personnel Economics: Competition in the workplace

Exercises

Exercise 8.1

Consider a profit-maximizing firm which employs two employees. Both employees must at least work from 9 am til 6 pm. Each employee earns a base salary of 50. The firm introduces a bonus B so as to induce workers to work longer hours. As the boss has difficulties observing how long workers actually work (the boss goes home at 6 pm himself), the bonus is conditional on a worker's performance. The worker who performs best will receive the bonus. When the workers perform equally well, each gets the bonus with probability $\frac{1}{2}$. Performance of a worker increases with his working hours. A worker will decide to work longer hours when the expected benefits of doing so are equal to or exceed the cost. For each employee, the cost of working longer hours are as follows:

Working from 6 pm til 8 pm cost 10

Working from 8 pm til 10 pm cost infinitely high

- a) Til what time will workers work when the bonus B is equal to 15?
- b) What is the minimal level of the bonus at which both workers choose working til 8?

Suppose an additional working hour increases the firm's revenues by 10.

- c) Will the firm implement a bonus?

Exercise 8.2

Consider a firm employing two workers and operating for two periods. In the first period, both workers earn salary W . In the second period, one of them is promoted and earns salary $B = W + Z$, whereas the other is retained and still earns W . Which worker is promoted depends partly on worker's effort and partly on luck. This is represented by the following function:

$$p_i = \frac{1}{2} + \pi(e_i - e_j)$$

where p_i is the probability that worker i is promoted, e_i is worker i 's effort in period 1, and e_j is worker j 's effort in period 1. Worker's cost of effort is given by $\frac{1}{2}\theta e^2$. Workers exert no effort in period 2.

- a) Determine the level of effort that each worker will exert in period 1. Interpret your result.

Suppose now that, with probability ρ , an outsider will get the job, and both workers earn W in period 2. With probability $1-\rho$, the promotion is assigned as described above.

- b) Determine the level of effort that each worker will exert in period 1. Interpret your result.

c) What is the probability of being promoted for each worker?

Rather than working at this firm, workers can decide to work somewhere else, and enjoy utility V in both periods.

d) Derive the level of the wage W for which workers are just willing to work at the firm. How does W depend on ρ ? Interpret your result.

Suppose $\rho=0$ and that a unit of effort is worth R to the firm.

e) Derive the profit-maximizing level of Z . Interpret your result.

Exercise 8.3 (adapted from Peter Kuhn's teaching material)

In several parts of this course, we have studied agent behavior and efficient contracts when the production process takes the form $Q = dE + \varepsilon$, where E is effort, d is a productivity parameter, and ε is a mean-zero random variable representing the role of luck. In doing so we generally assumed that the only action the agent can take is to choose a level of effort, so that higher levels of E lead to a higher expected value of output, Q . In some important contexts, however, agents can take a second type of action after the contract is signed. This action has no effect on the expected value of Q , but it increases the variance of Q , by increasing the variance of ε ; in other words this action increases the role of luck in the production process.

a) Explain (in words) why it might be in an agent's interest to take a costly action that increases the variance of ε .

b) Describe how this may apply to a fund manager in the financial industry who is rewarded based on his fund's relative performance.

Exercise 8.4 (taken from one of the last years' exams)

Astrid and Bella are sisters. Each of them has a job in which she can freely decide on her work hours. They earn w per hour. Astrid's utility function is: $wh_A + \mu(wh_A - wh_B) - (h_A)^2$ where the first term reflects utility from income and the last term reflects the utility costs from loss of leisure. The term in the middle reflects that Astrid finds it important how her income compares to the income of her sister. We assume that $\mu > 0$, and so assume that Astrid likes it when she earns more than her sister and dislikes it to earn less than her. The utility function of Bella is $wh_B + \mu(wh_B - wh_A) - (h_B)^2$. The sisters choose their hours independently from each other and simultaneously.

a) How many hours will Astrid and Bella choose to work?

b) Assume now that Astrid and Bella jointly decide on their work hours. They choose work hours so as to maximize the sum of their utilities. How many hours will Astrid and Bella work?

c) Calculate and compare the utility levels of Astrid and Bella under questions a) and b).

d) Now assume that Astrid and Bella can not jointly choose their work hours, but can jointly decide to implement an income tax scheme. Any tax receipts will be donated to a charity that Astrid and Bella do not care about. The income tax system can be as simple or complex as they like. Do Astrid and Bella jointly decide to implement a income tax scheme and, if yes, how does it look like? Motivate your answer.

Answers

8.1

a) The payoff matrix of the worker is:

	When the worker stays til 8, his payoff is:	When the worker leaves at 6, his payoff is:
Suppose other worker stays til 8	$\frac{1}{2}B - 10$	0
Suppose other worker leaves at 6	$B - 10$	$\frac{1}{2}B$

A bonus of 15 does not induce the worker to work til 8, irrespective of the choice of the other worker. The same holds for the other worker. And so, all workers go home at 6.

b) $B = 20$, see the matrix above.

c) Yes, extra cost = 20, extra revenues is $4 \times 10 = 40$.

8.2

a) $e = \pi Z / \Theta$

b) $e = (1-\rho)\pi Z / \Theta$

c) $\frac{1}{2}(1-\rho)$

d) $W + (1-\rho)\frac{1}{2}Z + W - \frac{1}{2}\Theta[(1-\rho)\pi Z / \Theta]^2 = 2V$. Rewrite: $W = V - \frac{1}{4}(1-\rho)Z + \frac{1}{4}(1-\rho)^2\pi^2 Z^2 / \Theta$.

$dW/d\rho = \frac{1}{4}Z - 2(1-\rho)\frac{1}{4}\pi^2 Z^2 / \Theta$. A higher ρ reduces the probability of getting the promotion, and so increases the wage W necessary to attract the worker. On the other hand, it also reduces effort, which permits a reduction in W . The total effect is ambiguous.

e) $Z = R/\pi$.

8.3

a) In general, there are two ways an agent can attain a 50% chance of winning a two-person tournament. One is by effort, i.e. by matching her opponent's effort level. The other is by luck, i.e. by increasing the role of luck in the contest so much that the winner is determined solely by luck. If the second action is less costly to the agent than the first, it will be in her interest to win this way. Note that is especially likely for agents who are behind in a contest, or for agents who have low ability because for them winning by effort may be either very costly or even impossible.

b) Brown, Harlow and Stark's study, which is discussed in the book, of mutual fund managers illustrates this idea. Fund managers who were behind their competitors at mid-year increased the riskiness of their portfolios in the hopes of doing well on some high-risk stocks and winning that way. Notably, they did this by actively selling safe stocks and buying low-risk stocks, not just by hanging on to stocks that had performed poorly.

8.4

a) Write down the utility function and differentiate to h_A and set equal to zero, which gives: $w + \mu w - 2h_A = 0$. Solving gives $h_A = \frac{1}{2} (1+\mu) w$. Likewise for h_B .

b) The sum of utilities is $wh_A + wh_B - (h_A)^2 - (h_B)^2$. Maximizing to h_A and h_B gives $w - 2h_A = 0$ and $w - 2h_B = 0$. Solving gives $h_A = h_B = \frac{1}{2} w$.

c) Astrid and Bella are best off if they decide to implement a tax rate of 100% for income above the income that corresponds with the number of hours that makes Astrid and Bella most happy, that is $h = \frac{1}{2}w$. The corresponding income is $\frac{1}{2}w * w = \frac{1}{2} w^2$. Income below this level remains untaxed.